

Reading a Quantum Paper: One-Page Cheat Sheet

From **How to Read a Quantum Paper** -- QGSS 2026 workshop, Olivia Lanes and James Weaver, IBM Quantum

1. Read in this order, not top-to-bottom

1. **Abstract** -- claimed contribution
2. **Conclusion** -- what authors believe they showed
3. **Figures + captions** -- the evidence, visually
4. **Introduction** -- context and motivation
5. **Methods + supplement** -- only as needed; caveats often hide here

Three questions to ask immediately

- What is the **claim**?
- What **evidence** supports it?
- What would **weaken** it?

2. Read each figure in this order

1. **Caption** first
2. **Axes, units, legend**
3. **Raw visual trend**
4. **Error bars / confidence regions**
5. **Fit lines vs measured points**
6. **Compare to the text's claim**

Captions and uncertainty matter more than color.

3. Then check for evidence quality

- ☐ **Authors and venue** -- who, where, what level of review
- ☐ **Uncertainty and baselines** -- error bars, comparisons, controls
- ☐ **Scope and caveats** -- where does the paper define its own limits?
- ☐ **Headline vs evidence** -- is the framing stronger than the data?

4. The 5-minute drill

After a first read of any paper, write down:

1. The **main claim** in one or two sentences
2. **Who** made it and **where** it appeared
3. One **figure-based** piece of evidence
4. One **scope-defining** detail (the precise version of the claim)
5. One reason for **skepticism** or **confidence**

5. Bold-claim red flags

- **No baseline named** -- "X% better than classical" without saying *which* method, on *which* problem
- **Engineering increment, breakthrough headline** -- "first N-qubit chip" with no gate fidelities
- **Speedup that hides the I/O** -- "exponentially faster" claims assuming data is already loaded
- **Application tied to a year** -- "drug discovery / break RSA / climate solved by 20XX"
- **Narrow result, broad headline** -- toy benchmark in paper, "real-world X solved" in press
- **Buzzword stack** -- quantum + AI + blockchain + agentic + ...
- **Press conference before peer review** -- announcement timed to marketing, not publication

6. Credibility ladder (strongest to weakest)

Peer-reviewed journal > arXiv preprint > conference talk > company blog > news headline / social post

The claim you repeat should be supported by the **strongest** source, not the loudest one.

No one reads a paper just once. Re-reading is normal, even for experienced readers.